

R Tutorial to fit unimodal spline method from the paper “Flexible unimodal density estimation in hidden Markov models”.

1. Setup

In this R tutorial, we illustrate the implementation of the unimodal spline method described in the paper “Flexible unimodal density estimation in hidden Markov models”. In this paper, we develop a method to fit hidden Markov models with flexible, P-spline-based emission densities that are constrained to be unimodal. We demonstrate how such constraints can make the model more robust against slight misspecification of the state process model which is often the case in practice.

We begin by first loading relevant packages and sourcing custom helper functions.

2. Understanding the mode location in unimodal splines

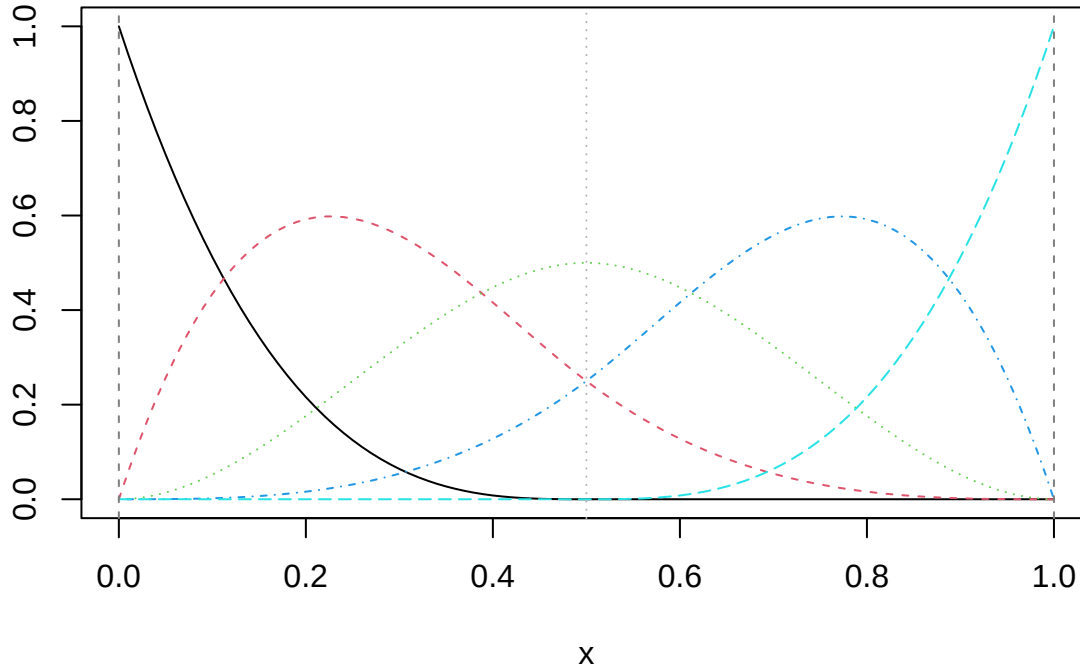
Let us first take a moment to think through how mode location of a B-spline and its coefficients interact. We start with a basic example. Suppose we have 5 B-spline basis functions defined on $[0,1]$, which means we also have 5 corresponding spline coefficients, denoted as $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$. These basis functions are typically defined by a set of knots, and for simplicity, let’s assume just one interior knot at $x = 0.5$. This knot splits the domain $[0,1]$ into two intervals:

- $[0,0.5)$
- $[0.5,1]$

Let’s take a look at the 5 basis functions.

```
plot(bsMat, mark_knots = "all")
title("B-spline basis functions")
```

B-spline basis functions



As we described in the main manuscript, to enforce unimodality (i.e., the fact that the spline should have a single peak/ mode), we need to apply some restrictions on the spline coefficients $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 . We fix the general shape by requiring the coefficients to be non-decreasing up to the mode and then non-increasing (Köllmann, Bornkamp, and Ickstadt 2014). In practice, the α parameters are constrained to sum to one and we apply the constraint to the unconstrained coefficients β . For simplicity, we will focus on the α values here, just keep in mind that later the constraint will be applied to the unconstrained coefficients.

Because unimodality is enforced via constraints on the spline coefficients, and not directly on the function values or the location of the mode, we cannot predict in advance exactly where the mode of the spline function will be on the x-axis because the basis functions overlap. Thus a single coefficient doesn't control the function height at a single point on the x-axis, but rather influences the entire region where the associated basis function is non-zero. So even if we have an increasing then decreasing sequence of spline coefficients, the actual mode of the function might fall in the following locations: (1) In the first interval, (2) at the knot ($x = 0.5$), or (3) in the second interval, depending on the interaction of the coefficients.

Let us see how those three locations look like if we consider an increasing then decreasing sequence of spline coefficients.

```
# Define increasing then decreasing sequence of spline coefficients:
alpha1 <- c(0.1, 2, 10, 9, 2) # mode is in the second interval
spline1 <- bsMat %*% alpha1
alpha2 <- c(1, 2, 10, 2, 1) # the mode is at the knot
spline2 <- bsMat %*% alpha2
alpha3 <- c(2, 9, 10, 2, 0.1) # mode is in the first interval
spline3 <- bsMat %*% alpha3

par(mar = c(5, 4, 4, 8), xpd = TRUE) # xpd=TRUE allows drawing outside plot

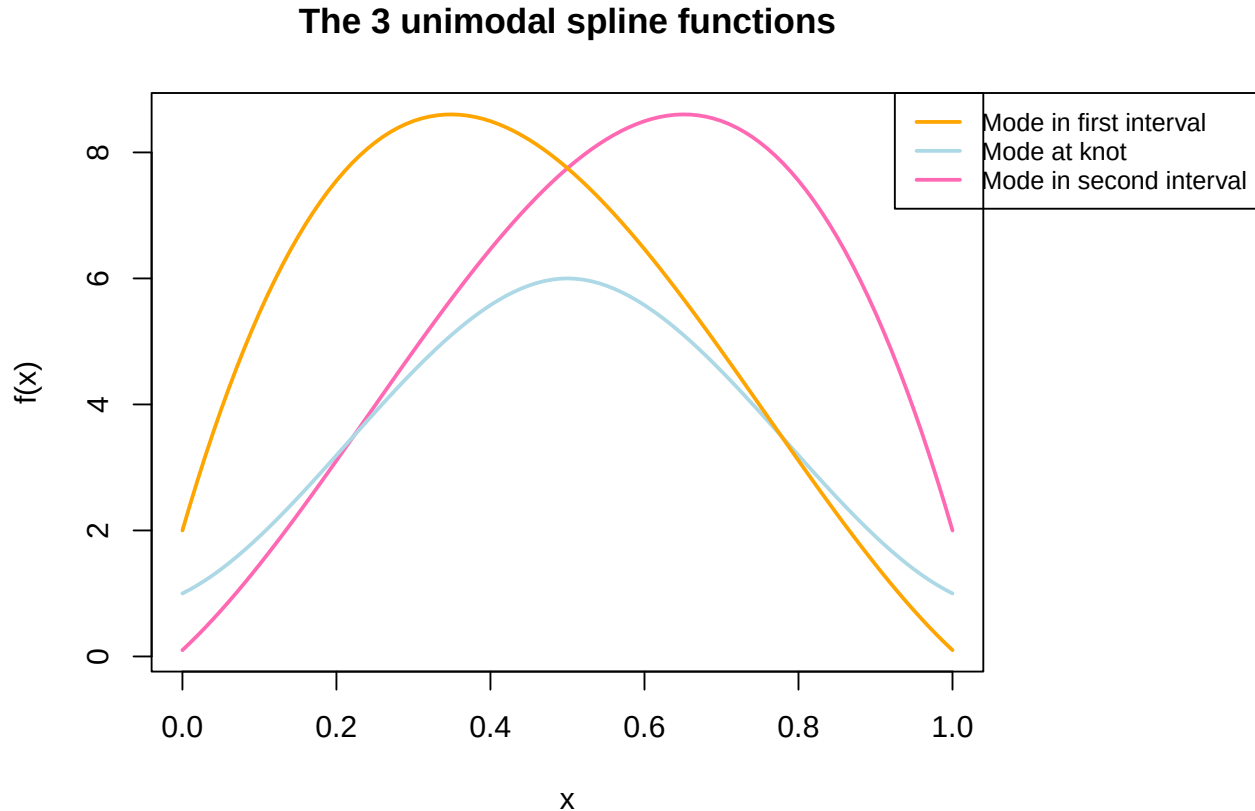
# Plot the functions
plot(x, spline1, type = "l", ylab = "f(x)", col = "hotpink", lwd = 2)
lines(x, spline2, col = "lightblue", lwd = 2)
```

```

lines(x, spline3, col = "orange", lwd = 2)
title("The 3 unimodal spline functions")

# Add the legend outside the plot
legend("topright", inset = c(-0.33, 0), # negative x inset pushes legend outside
      legend = c("Mode in first interval", "Mode at knot", "Mode in second interval"),
      col = c("orange", "lightblue", "hotpink"),
      lwd = 2,
      cex = 0.8)

```



Observe that in all three cases, the peak value of the sequence $\alpha_1, \dots, \alpha_5$ occurs at the third coefficient. However, the mode of the function $f(x)$ (i.e., the location in the x -axis where the function reaches its maximum) differs across the three examples. This illustrates an important point: the location of the peak in $f(x)$ is influenced not only by where the α 's peak is in the sequence, but also by the shape and placement of the B-spline basis functions and the specific value of all coefficients. In other words, even with the same unimodality constraint on the coefficients, the interaction between the α 's and the basis functions can shift the location of the mode in the function output. This is **key** when interpreting or designing unimodal spline models.

Observe that we only considered the “conventional” unimodality definition as increasing then decreasing sequence. However, unimodality can also be achieved with other valid configurations. For example:

- constant then decreasing
- increasing then constant

These alternative shapes still satisfy the definition of unimodality — having a single peak — but allow for plateau or shoulders rather than a sharp maximum. Below, we illustrate these two alternative cases:

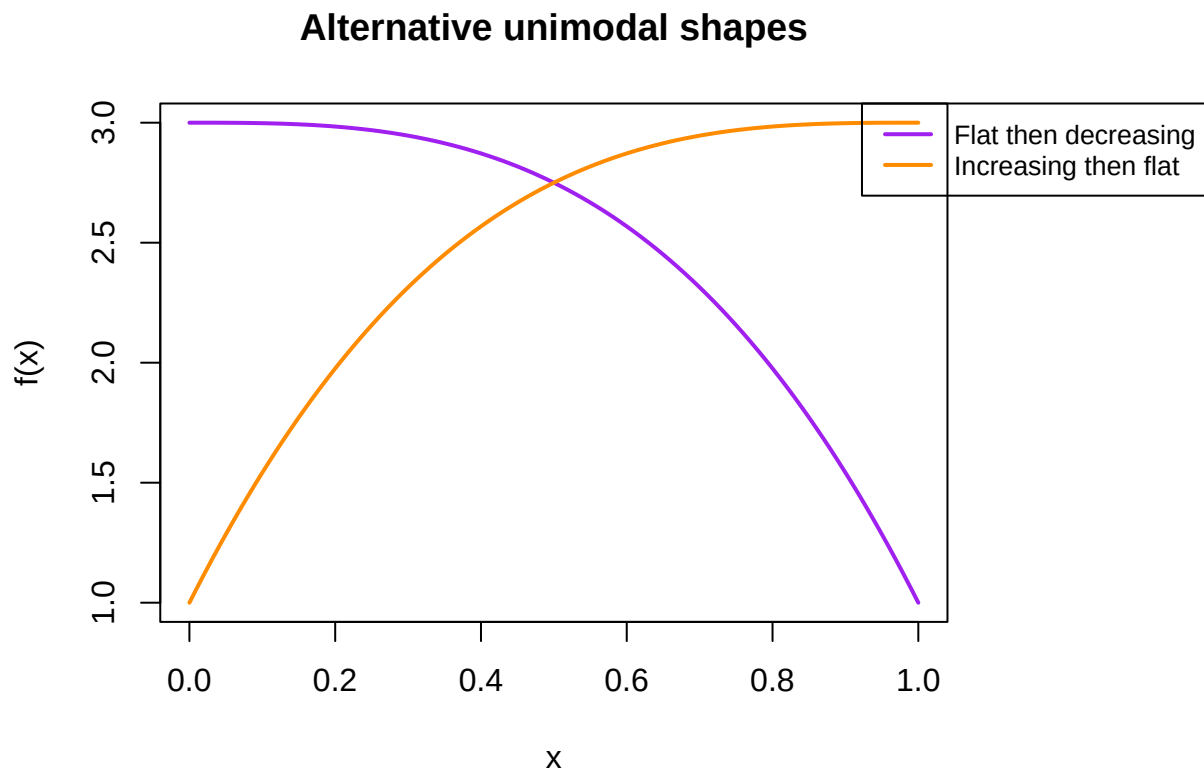
```

alpha4 <- c(3, 3, 3, 2, 1) # flat then decreasing
spline4 <- bsMat %*% alpha4

alpha5 <- c(1, 2, 3, 3, 3) # increasing then flat
spline5 <- bsMat %*% alpha5

par(mar = c(5, 4, 4, 8), xpd = TRUE)
plot(x, spline4, type = "l", ylab = "f(x)", col = "purple", lwd = 2)
lines(x, spline5, col = "darkorange", lwd = 2)
title("Alternative unimodal shapes")
legend("topright", inset = c(-0.33, 0),
      legend = c("Flat then decreasing", "Increasing then flat"),
      col = c("purple", "darkorange"),
      lwd = 2,
      cex = 0.8)

```



3. Unimodal splines for HMMs

Now that we have a better intuition for the unimodality constraint and the mode's location, we can fit unimodal splines to a dataset. To demonstrate the application of nonparametric unimodal estimation for state-dependent densities within HMMs, we will use the `nessi` dataset and the `LaMa` R package. For detailed instructions on how to use the `LaMa` package to fit nonparametric densities, refer to the vignette available at [LaMa vignettes on Penalised Splines](#). We will work on acceleration data of the Loch Ness Monster. More specifically, we will work on the overall dynamic body acceleration (ODBA) datastream.

```
head(nessi)
```

```
##          ODBA    logODBA state
```

```
## 1 0.03775025 -3.276763      2
## 2 0.05417830 -2.915475      2
## 3 0.03625247 -3.317248      2
## 4 0.01310802 -4.334531      1
## 5 0.05402441 -2.918319      3
## 6 0.06133794 -2.791357      3
```

The dataset contains three columns: the overall dynamic body acceleration (ODBA) of `nessi`, `logODBA`, which represents the logarithm of ODBA and is our variable of interest, and the `state` column, which indicates the behavioural state of the monster. The `state` ranges from 1 to 3, as `nessi` was simulated with a three-state hidden Markov model (HMM). Since ODBA is strictly positive with some extreme values, direct analysis can be difficult. Therefore, for our analysis, we consider the logarithm of ODBA as our observed process. We will fit a three-state HMM with unimodal nonparametric state-dependent distributions.

First, let us specify the model we will fit to our data. To do so, we use the `smooth_dens_construct()` function, available in `LaMa`.

We set the following arguments:

- `"data"`: dataframe containing the variables used in the formula. It can be multiple data streams of different types (`real`, `positive`, `circular`). This information is then converted into a suitable B-spline basis.
- `"par"`: nested named list of initial means and sds/concentrations for each data stream. This information is converted into initial guesses for the spline coefficients to facilitate easier numerical estimation.
- `"type"`: For each data stream, either “real” for data on the reals, “positive” for data on the positive reals or “circular” for angular data. Needs to be a vector corresponding to the number of data streams in `data`.
- `"k"`: number of basis functions for each data stream
- `"knots"`: optional named list of knots vectors (including the boundary knots) to be used for basis construction. If not provided, the knots are placed equidistantly for “real” and “circular” and using polynomial spacing for “positive”.

`smooth_dens_construc()` then builds the P-spline design and penalty matrices for each data stream as well as a matrix of initial spline coefficients based on the provided parameters. The basis functions are standardised such that they integrate to one, which is crucial for density estimation.

```
## logODBA
## Leaving out last column of the penalty matrix, fix the last spline coefficient at zero for identifiability
## Parameter matrix excludes the last column. Fix this column at zero!
```

We can now have a look at what `smooth_dens_construc()` returns: a nested list containing the design matrices `Z`, the penalty matrices `S`, the initial coefficients `coef`, the prediction design matrices `Z_predict`, the prediction grids `xseq`, and details for the basis expansion for each data stream. More details about `smooth_dens_construct()` are available in the function documentation.

```
# objects for model fitting
Z <- modmat$Z$logODBA # spline design matrix for logODBA
S <- modmat$S$logODBA # penalty matrix for logODBA
beta <- modmat$coef$logODBA # initial spline coefficients matrix of dimension N by k-1
w <- modmat$basis$logODBA$w # 1 / integral of the basis functions,
# needed for reweighting in the unimodality penalty (see paper)
```

```

# objects for prediction / plotting
Z_pred <- modmat$Z_predict$logODBA # prediction design matrix
xseq <- modmat$xseq$logODBA # prediction sequence of logODBA values
N <- 3 # number of states

```

Before minimising the negative penalised log-likelihood, we need to define the initial parameter list `par0` and a `dat` list that contains the data and potential hyperparameters. The elements in the `dat` list are not estimated by the optimizer. Note that we specified `m` as the indices where the sign of the spline coefficients should change, for each state. This choice was based on preliminary analysis. In general, the mode position should always be explored, which we will address in the next section.

```

# coefficient mode position for each state
m <- c(8, 12, 15) # one index per state, index <= 25

# initialise parameters
par0 <- list(beta0 = rep(-2, 6), # initial transition matrix on logit scale
             beta = beta) # spline coefficients prepared by buildSmoothDens()

# initialise data and hyperparameters
mydat <- list(N = N, # number of states
              Z = Z, # spline design matrix
              S = S, # spline penalty matrix
              lambda = rep(10, 3), # initial smoothness penalty parameters
              m = m, # coefficient mode indices
              kappa = 1e4, # strength parameter of the unimodality penalty
              logweights = t(log(w[-k] / w[k]))[rep(1, N), ]) # weights

```

Observe that `beta` has 24 columns, instead of 25. This is because the last column, which corresponds to the last coefficient for each state, needs to be fixed to zero for identifiability which we do line 216 by using `cbind(beta, 0)`. Then, we transform the unconstrained parameter matrix to non-negative weights that sum to one (α in the main manuscript) for each state using the inverse multinomial logistic link. The columns of the `allprobs` matrix are then computed as linear combinations of the columns of `Z` and the weights `alpha`. We then penalise the unconstrained coefficients `beta` using the `penalty()` function. This helps prevent the estimated densities from overfitting by becoming too wiggly. Finally, we apply the unimodality constraint on the `betas` using the `penalty_uni()` function. Note that we place the unimodality penalty on the reweighted coefficients `beta + logweights`. This is because the penalty needs to be on the scale of the original B-spline basis, which was standardised to integrate to one by `smooth_dens_construct()`. The reweighting is done to ensure that the unimodality constraint is applied correctly, as described in the main manuscript (Equation (5)).

```

pnll = function(par){
  getAll(par, dat)
  ## state process model
  Gamma <- tpm(beta0)
  delta <- stationary(Gamma)
  ## state-dependent process model
  # multinomial logit for density weights
  alpha <- exp(cbind(beta, 0)) # last column is fixed at zero
  alpha <- alpha / rowSums(alpha); REPORT(alpha) # reporting for easy access
  # state-dependent densities
  allprobs <- matrix(1, nrow(Z), N)
  ind <- which(!is.na(Z[,1])) # NA handling

```

```

# now just linear combinations of precomputed basis functions in Z
allprobs[ind,] <- Z[ind,] %*% t(alpha) # this does Z %*% alpha[i,] for all states at once
# forward algorithm + smoothness penalty + unimodality penalty
- forward(delta, Gamma, allprobs) +
  penalty(beta, S, lambda) +
  penalty_uni(beta + logweights, m, kappa) # reweighting based on eq (5) in the main manuscript
}

```

We jointly estimate the model coefficients and the optimal degree of smoothness using the `qrem1()` function which an approximate restricted maximum likelihood procedure (Koslik 2024). See the LaMa vignette on Penalised Splines for more details.

Now let us have a look at the model output.

```

# Get the estimated (constrained) spline coefficients
Alphas = mod$alpha

# Compute the state-dependent densities on prediction grid
Dens = Z_pred %*% t(Alphas)

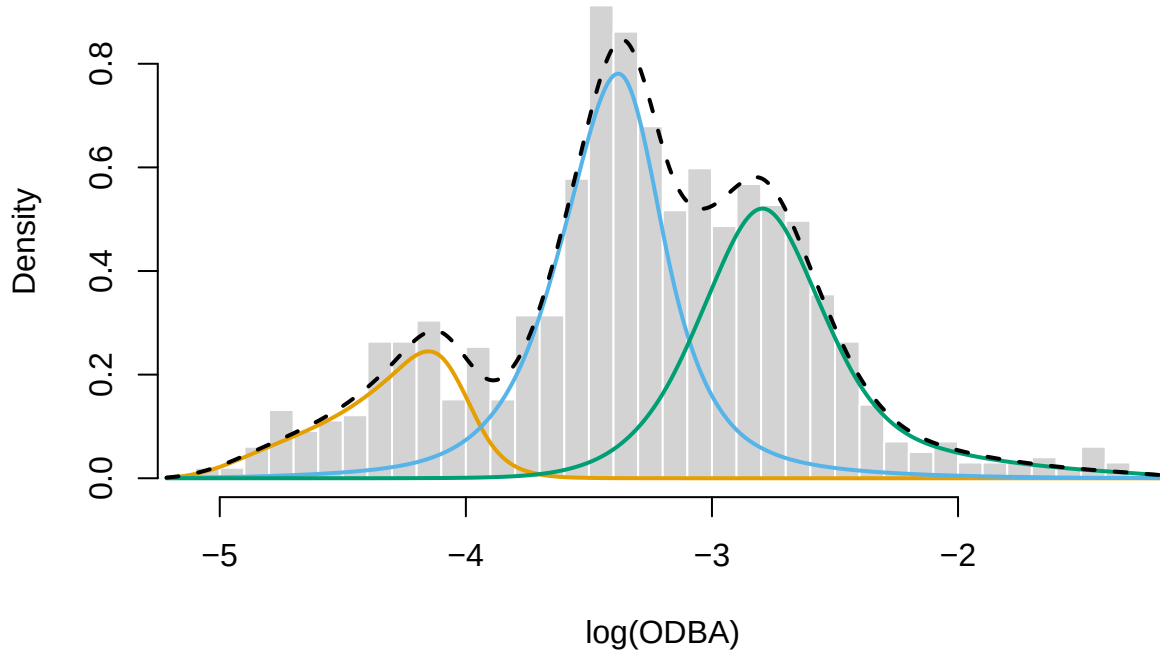
# Get the stationary distribution
Deltas = mod$delta

# Compute marginal density
marDens = colSums(t(Dens) * Deltas)

color = c("#E69F00", "#56B4E9", "#009E73")
par(mfrow = c(1,1))
hist(data$logODBA, prob = TRUE, breaks = 50, bor = "white",
      main = "Estimated state-dependent densities", xlab = "log(ODBA)")
for(i in 1:N){
  lines(xseq, Deltas[i] * Dens[,i], col = color[i], lwd = 2)
}
lines(xseq, marDens, lty = 2, lwd = 2)

```

Estimated state-dependent densities



The unimodality constraint is implemented in the last line of `pnll` by calling the function `penalty_uni()`. Internally, this function applies the following strategy, described in the manuscript. The linear positivity constraint is written as

$$C'\beta \geq 0.$$

Instead of enforcing this as a hard constraint, we incorporate it as a penalty in the objective function by adding

$$\min(C'\beta, 0),$$

scaled by a large penalty parameter (called `kappa` in `penalty_uni()`) to strongly discourage violations. Since the standard minimum function is not differentiable at 0, we use a smooth approximation, `min0_smooth()`, to ensure continuity and differentiability in the optimization process. Parameter `rho` (defaulting to 20) controls the smoothness of the approximation: the larger, the closer to the minimum function as we can see in the plot below.

```
# Sequence of x values
x_vals <- seq(-1, 1, length.out = 400)

# List of alpha values to try
rhos <- c(1, 5, 10, 20, 50, 100)

# Set up the plot
plot(x_vals, min0_smooth(x_vals, rho = rhos[1]), type = "l",
     ylim = c(-2, 0), lwd = 2, col = 1,
     xlab = "x", ylab = "min0_smooth(x)",
     main = "Smooth approximation of min(x, 0) for different rho values")

# Add curves for other alpha values
```

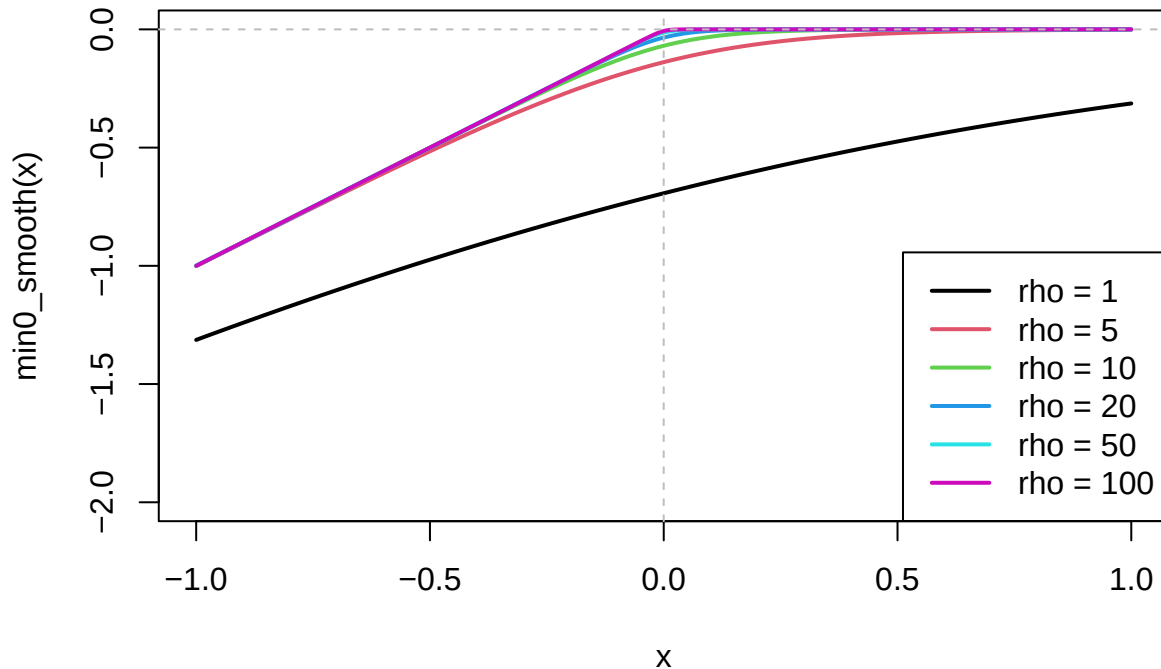
```

for (i in 2:length(rhos)) {
  lines(x_vals, min0_smooth(x_vals, rho = rhos[i]), col = i, lwd = 2)
}

# Add legend
legend("bottomright", legend = paste("rho =", rhos), col = 1:length(rhos), lwd = 2)
abline(h = 0, v = 0, col = "gray", lty = 2)

```

Smooth approximation of $\min(x, 0)$ for different rho values



Now we are ready to fit the whole model and search for the mode instead of specifying it subjectively!

4. Search for the mode

Using the fit with the selected mode m , we can identify potential indices for the change of sign in the unimodality constraint, which allows us to avoid exploring the entire range of possible modes. We assign the states in increasing order: the first state corresponds to the leftmost mode, the second to the central mode, and the third to the rightmost mode.

```

# pick candidates manually
m1 <- 5:8   # Potential constraints indices for the first state
m2 <- 10:13 # Potential constraints indices for the second state
m3 <- 14:16 # Potential constraints indices for the third state

```

Then we can build the grid of exploration and set up the model as before.

```

m_grid <- expand.grid(m1 = m1, m2 = m2, m3 = m3)

mydat <- list(N = N, # number of states
             Z = Z, # spline design matrix

```

```

S = S, # spline penalty matrix
lambda = rep(10, 3), # initial smoothness penalty parameters
kappa = 1e4,
logweights = t(log(w[-k] / w[k]))[rep(1, N), ] # weights

```

We can then fit a model for each combination of indices. In total, this involves fitting $\text{length}(m1) \times \text{length}(m2) \times \text{length}(m3)$ models, which amounts to 48 in our case. For convenience, we currently explore only one randomly selected initial value `Par0` and `modmat$par`, but ideally, we should examine more, meaning even more models to fit.

```

llks = rep(NA, nrow(m_grid))
for(i in 1:nrow(m_grid)){
  mydat$m = m_grid[i, ]
  thismod = qrem1(pnll, par0, mydat, random = "beta", silent = 2)
  llks[i] = thismod$llk
}

```

```

# Get the mode estimate
m_opt = m_grid[which.max(llks), ] # m_opt is 7 12 15

# Update constraint matrix with the optimal mode
mydat$m = m_opt
mydat$kappa = 1e3 * 2

# Update parameters using the previous model fit
par0$beta = mod$par0$beta
mydat$lambda = mod$lambda * 2

# Fit the model
mod_c = qrem1(pnll, par0, mydat, random = "beta")

```

```

## Creating AD function
## Initialising with lambda: 1.351 7.271 12.42
## outer 1 - lambda: 1.256 4.04 7.797
## outer 2 - lambda: 1.196 3.469 6.321
## outer 3 - lambda: 1.163 3.315 5.843
## outer 4 - lambda: 1.142 3.273 5.687
## outer 5 - lambda: 1.131 3.261 5.638
## outer 6 - lambda: 1.125 3.258 5.62
## outer 7 - lambda: 1.121 3.257 5.615
## outer 8 - lambda: 1.12 3.257 5.613
## outer 9 - lambda: 1.119 3.257 5.613
## Converged
## Final model fit with lambda: 1.119 3.257 5.613

```

```

# Get the results
alpha_c = mod_c$alpha
delta_c = mod_c$delta
dens_c = t(delta_c * t(Z_pred %*% t(alpha_c)))

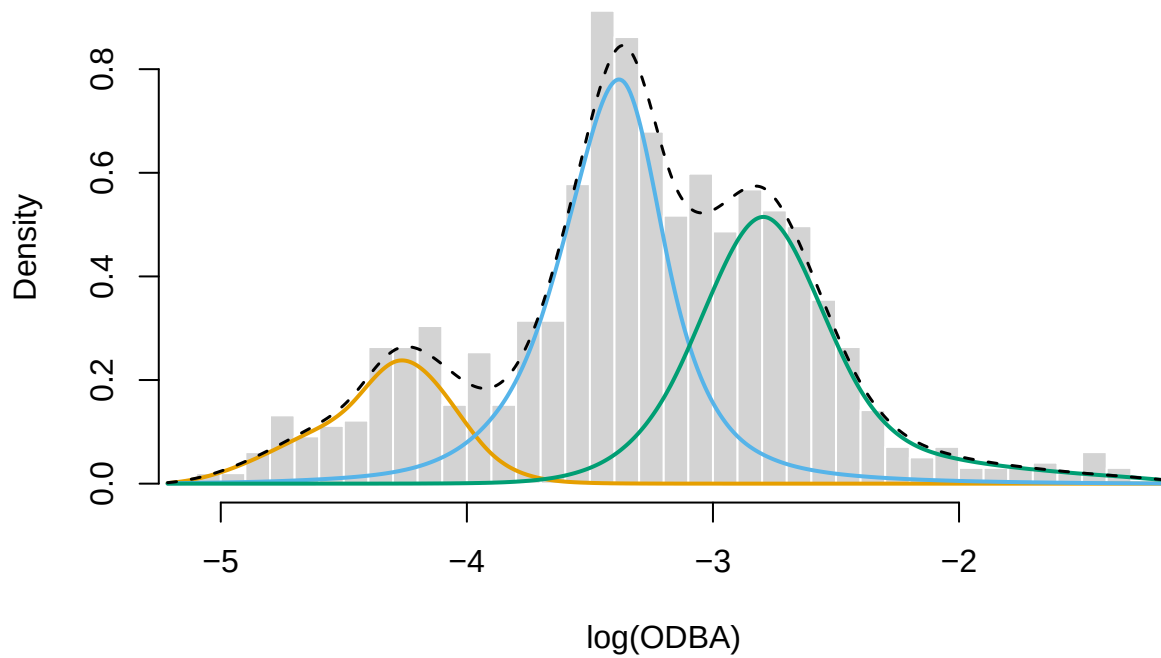
```

Now, we can have a look at our final estimated model

```
# state-dependent and marginal distributions
par(mfrow = c(1,1))

hist(data$logODBA, breaks = 30, prob = TRUE, bor = "white",
     main = "Unimodal density estimates",
     xlab = "log(ODBA)", ylab = "Density")
for(i in 1:3) lines(xseq, dens_c[,i], col = color[i], lwd = 2)
lines(xseq, rowSums(dens_c), lty = 2, lwd = 1.5)
```

Unimodal density estimates



References

- Köllmann, Claudia, Björn Bornkamp, and Katja Ickstadt. 2014. "Unimodal Regression Using Bernstein–Schoenberg Splines and Penalties." *Biometrics* 70 (4): 783–93.
- Koslik, Jan-Ole. 2024. "Efficient Smoothness Selection for Nonparametric Markov-Switching Models via Quasi Restricted Maximum Likelihood." *arXiv Preprint arXiv:2411.11498*.